Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Algorithms Laboratory		
Academic Year	2025–2026 (Even)	Batch	2023–2027
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Due Date	27.01.2026		

Experiment 2: Binary Classification using Naïve Bayes and K-Nearest Neighbors

1. Aim and Objective

To implement Naïve Bayes and K-Nearest Neighbors (KNN) classifiers for a binary classification problem, evaluate them using multiple performance metrics, visualize model behavior, and analyze overfitting, underfitting, and bias–variance characteristics.

2. Dataset Description

The Spambase dataset is a benchmark binary classification dataset used to identify spam emails. It consists of numerical features extracted from email content and a binary class label indicating spam or non-spam.

Dataset Reference:

- Kaggle – Spambase Dataset

3. Preprocessing Steps

- The dataset was loaded into a Pandas DataFrame for preprocessing.
- Input features and target labels were separated for modeling.
- A stratified train–test split was performed to preserve class distribution.
- Feature scaling was applied using the StandardScaler technique.

4. Implementation Details

- Implemented Gaussian, Multinomial, and Bernoulli Naïve Bayes classifiers
- Implemented baseline KNN classifier
- Tuned KNN hyperparameters using GridSearchCV and RandomizedSearchCV
- Compared KDTree and BallTree neighbor search algorithms

5. Visualizations

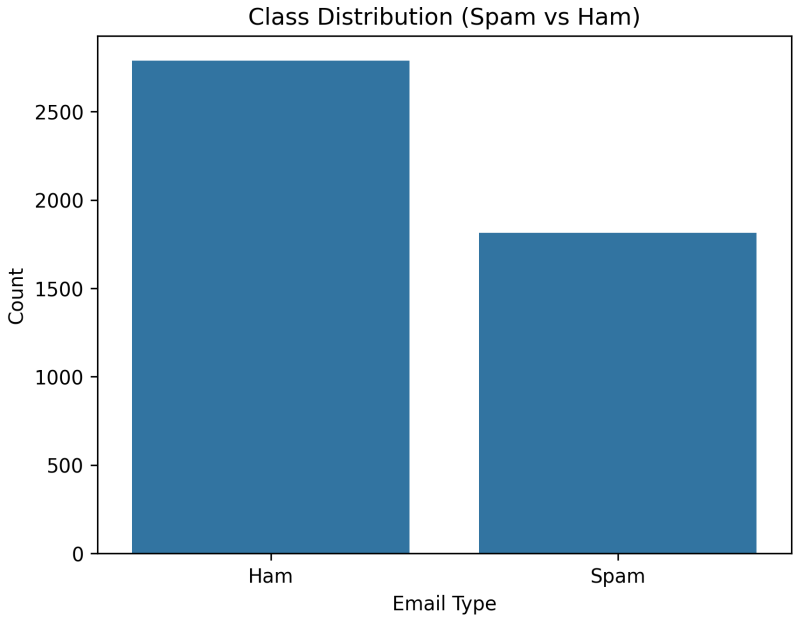

Figure 1: Class Distribution of Spam and Non-Spam Emails

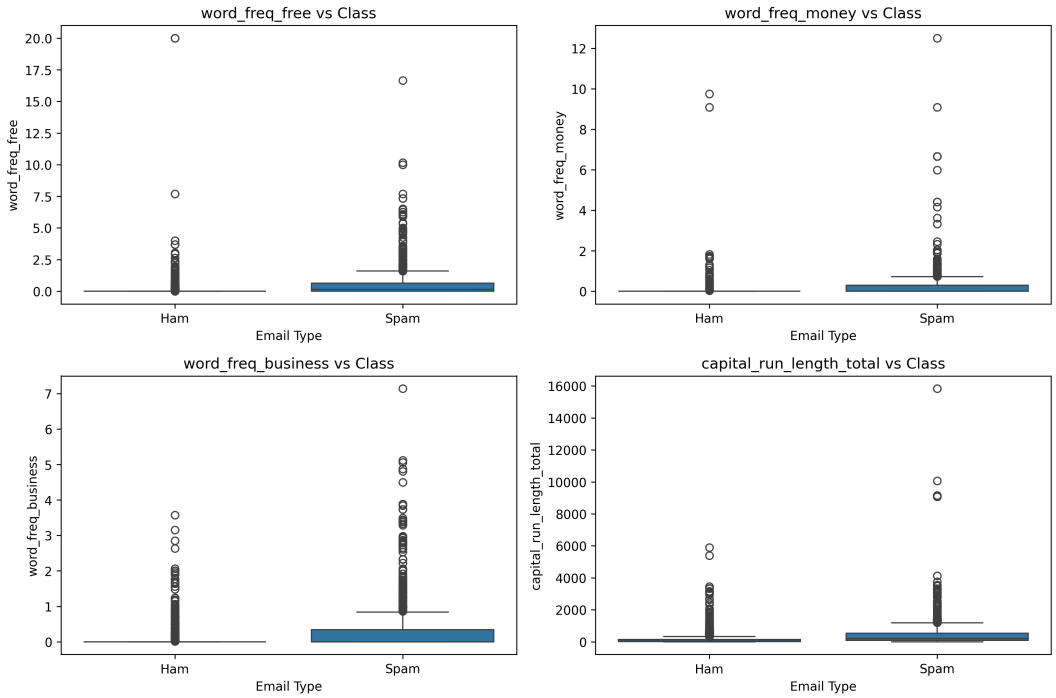

Figure 2: Box Plots of Important Features

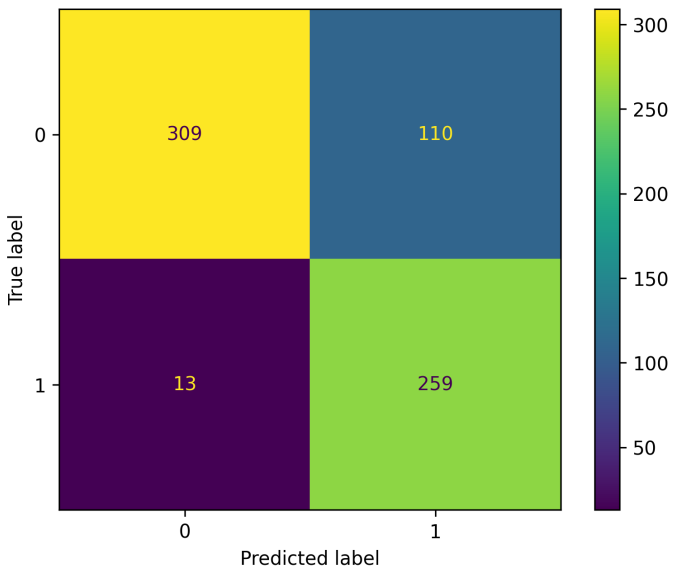

Figure 3: Confusion Matrix for Gaussian Naïve Bayes

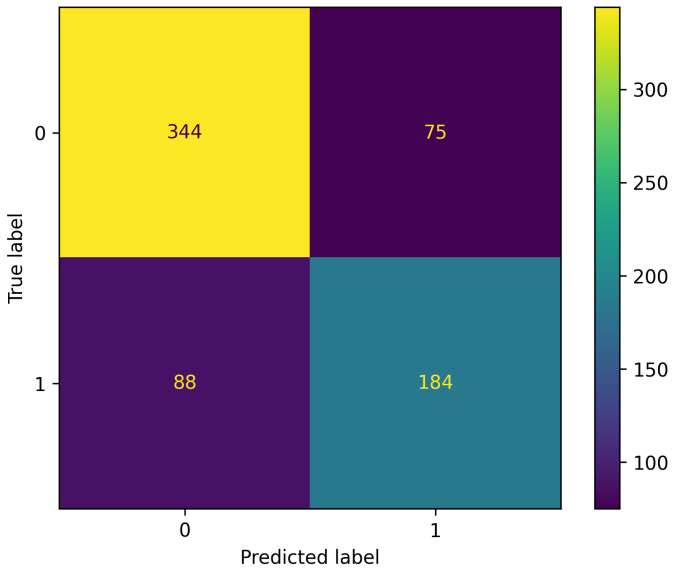

Figure 4: Confusion Matrix for Multinomial Naïve Bayes

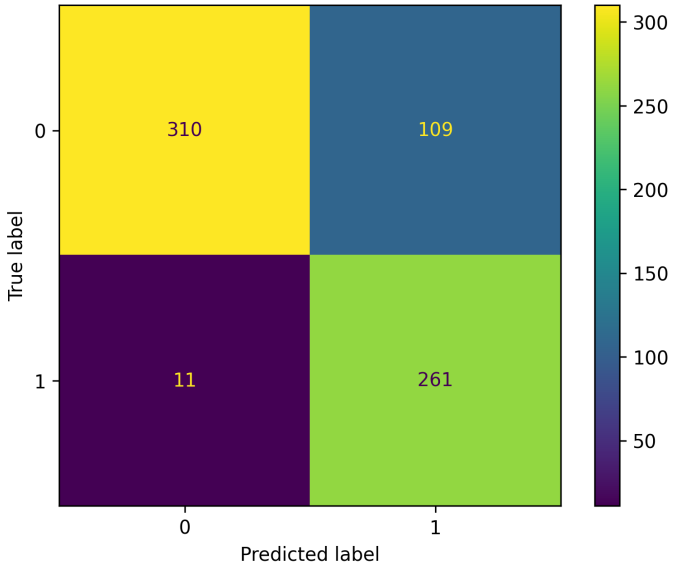

Figure 5: Confusion Matrix for Bernoulli Naïve Bayes

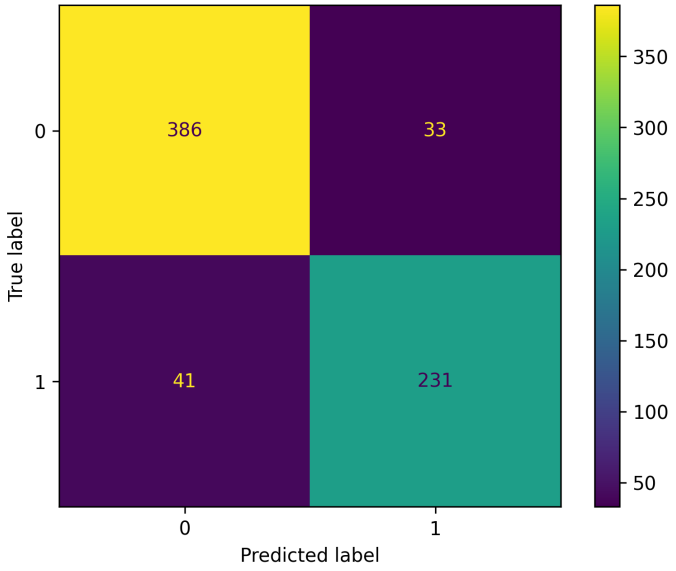

Figure 6: Confusion Matrix for KNN (K = 5)

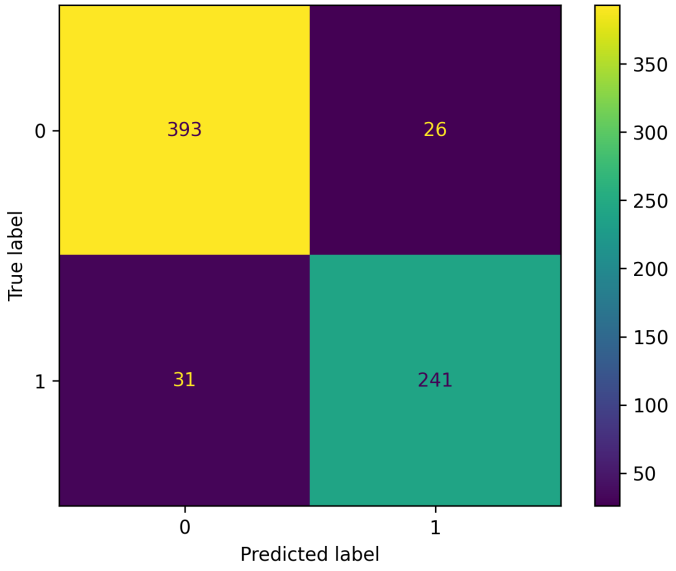

Figure 7: Confusion Matrix for Randomized KNN

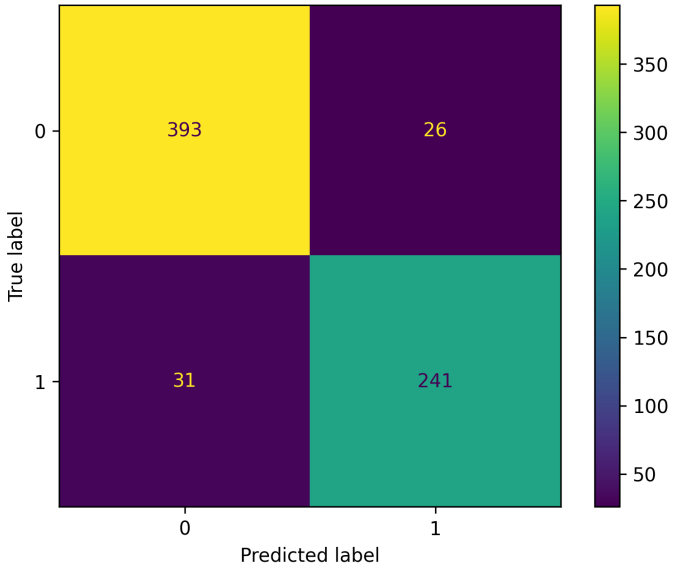

Figure 8: Confusion Matrix for KD Tree

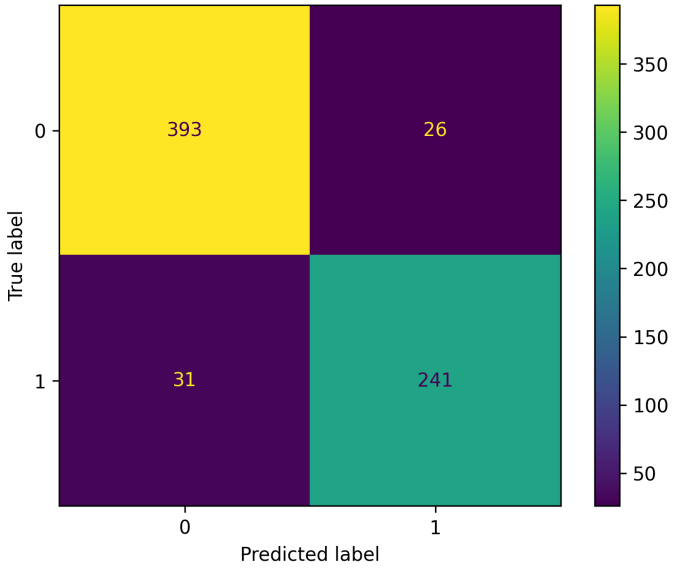

Figure 9: Confusion Matrix for Ball Tree

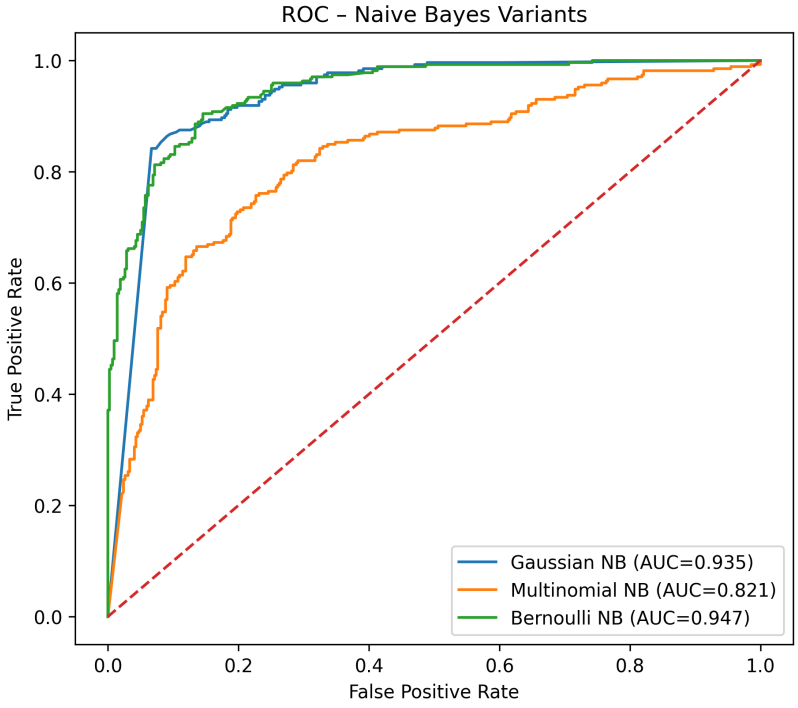

Figure 10: ROC Curve for Naive Bayes Variants

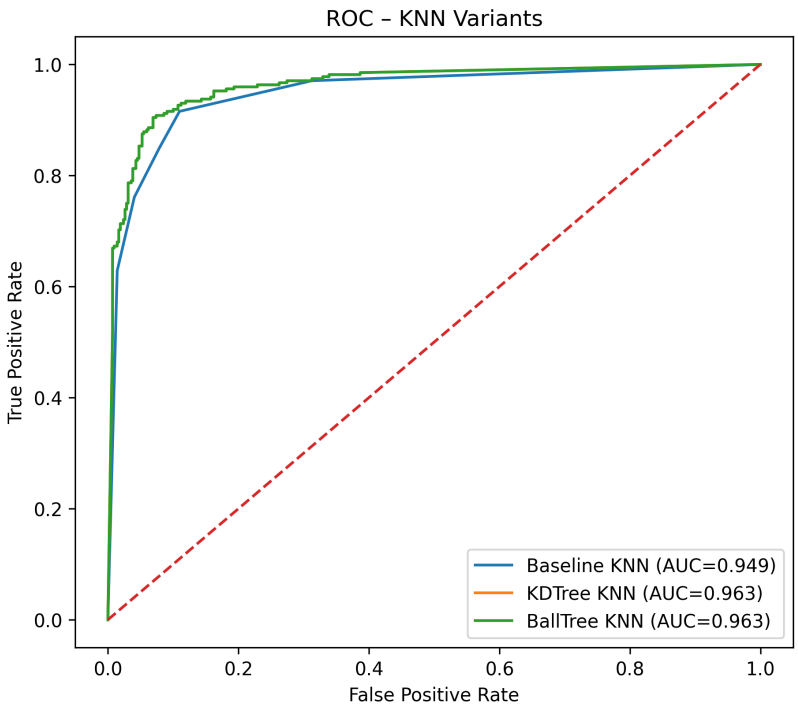

Figure 11: ROC Curve for KNN Variants

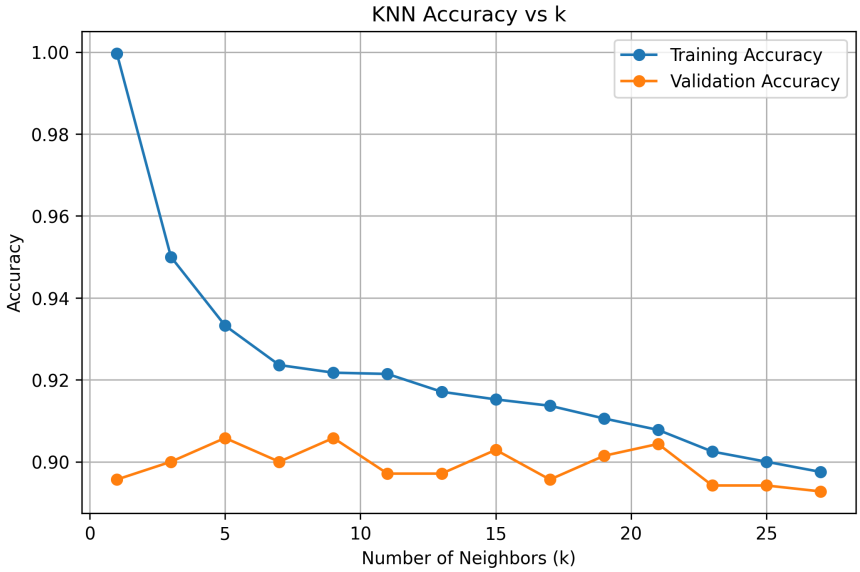

Figure 12: Accuracy vs. k Plot for KNN

6. Performance Tables

Table 1: Naïve Bayes Performance Metrics

Metric	**Gaussian NB**	**Multinomial NB**	**Bernoulli NB**
Accuracy	0.8219	0.7641	0.8263
Precision	0.7018	0.7054	0.7104
Recall	0.9522	0.6764	0.9595
F1 Score	0.8081	0.6930	0.8130
Specificity	0.7374	0.8210	0.7398
Training Time (s)	0.0101	0.005	0.008

Table 2: KNN Hyperparameter Tuning

Search Method	**Best Parameters**	**Best CV Accuracy**
Grid Search	k=9, distance, base KNN	0.9183
Randomized Search	k=10, distance, base KNN	0.9170

Table 3: KNN Performance using KDTree

Metric	**Value**
Optimal k	9
Accuracy	0.9175
Precision	0.9026
Recall	0.8860
F1 Score	0.8942
Training Time (s)	0.0426
Prediction Time (s)	0.1635

Table 4: KNN Performance using BallTree

Metric	**Value**
Optimal k	9
Accuracy	0.9175
Precision	0.9026
Recall	0.8860
F1 Score	0.8942
Training Time (s)	0.020
Prediction Time (s)	0.1658

Table 5: Comparison of Neighbor Search Algorithms

Criterion	**KDTree**	**BallTree**
Accuracy	0.9175	0.9175
Training Time (s)	0.0426	0.020
Prediction Time (s)	0.1635	0.1658
Memory Usage	Low / Medium	Medium / High

7. Overfitting and Underfitting Analysis

- **Difference Between Training and Validation Accuracy**
  - A large gap between training and validation accuracy indicates high variance and overfitting.
  - Similar values on both sets suggest good generalization and low variance.
- **Effect of Small and Large Values of k**
  - Small k values lead to low bias but high variance, making the model sensitive to noise.
  - Large k values increase bias by oversmoothing decision boundaries.
- **Role of Hyperparameter Tuning in Generalization**
  - Hyperparameter tuning controls model complexity and influences the bias–variance trade-off.
  - Appropriate tuning improves generalization, whereas overly aggressive tuning can cause overfitting.

8. Bias–Variance Analysis

- **Bias Behavior of Naïve Bayes**
  - Strong feature-independence assumptions limit model flexibility.
  - Leads to generally high bias and low variance, as reflected by similar training and validation performance.
- **Variance Behavior of KNN**
  - Model complexity depends on the neighborhood size k.
  - Small k values result in low bias but high variance.
- **Effect of Tuning on the Bias–Variance Trade-off**
  - Hyperparameter tuning modifies model complexity.
  - Aggressive tuning in KNN reduced bias but increased variance, causing overfitting.

9. Observations and Conclusion

Compared several classification models using training and validation metrics to study their generalization behavior. The experiments show that model complexity and hyperparameter tuning significantly affect performance, with simpler models offering stable but moderate results and more complex methods requiring careful control to avoid overfitting. Cross-validation was essential for fair comparison and parameter selection. Overall, the study emphasizes the importance of balancing accuracy and robustness when selecting machine learning models.

References

- Scikit-learn – Naïve Bayes Documentation
- Scikit-learn – KNN Documentation
- Scikit-learn – Hyperparameter Optimization
- Kaggle – Spambase Dataset